The D. G. Ruparel College
OF ARTS, SCIENCE AND COMMERCE
Department of Computer Science • Information Technology

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TITLE :

Aim:- Write a program to read the CPU%, Memory %, and CPU temperature of Raspberry Pi.

* Hardware Requirements
1. Raspberry Pi Model A/B/B+

* Software Requirements
1. Raspbian stretch OS

* Code:-

```
#!/usr/bin/env python3

import psutil
import time

def get_cpu_temperature():
    """Read Raspberry Pi CPU temperature."""
    try:
        with open("/sys/class/thermal/thermal_zone0/temp
                  ", "r") as file:
            temp = int(file.read()) / 1000.0
```

```
        return temp
    except FileNotFoundError:
        return None

while True:
    # CPU usage percentage
    cpu_usage = psutil.cpu_percent(interval=1)

    # Memory usage percentage
    memory_usage = psutil.virtual_memory().
                                    percentnt

    # CPU temperature
    cpu_temp = get_cpu_temperature()

    print("-------------------")
    print(f"CPU Usage : {cpu_usage:.1f}% ")
    print(f"Memory Usage : {memory_usage:.1f}%")

    if cpu_temp is not None:
        print(f"CPU Temperature : {cpu_temp:.1f}
                                        °C")
    else
        print("CPU Temperature : Not available")
    print("-------------------")
    time.sleep(1)
```

To Run the Programme :-

1. sudo apt update
2. sudo apt install python3-psutil
3. python3 test.py